

energy technologies, and we are now off to take a quick look at some of these.

Electronics and solar power, an obvious synergy

"The most feasible green energy technologies available in India are hydro, wind, biomass and solar. The first three are already being harnessed at various levels of their potential but there is a huge opportunity for solar energy in India as we receive about 3000 hours of sunshine every year—equivalent to over 5000 trillion kWh! We can utilise this solar energy to provide a clean source of continuous electricity to the entire nation. We strongly believe that the scarcity of fossil fuels and their effect on our environment will draw more attention to renewable energy, especially solar energy. We should not forget that the sun is a source of infinite energy and it generates more energy than what the world consumes," says Hari Kiran, managing director, Sujana Energy.

Solar panels are no longer confined to bulky boilers on rooftops. Thin-film solar panels and cells that are lightweight, flexible and efficient have made it possible to incorporate solar energy, as a backup if not as a primary power source, in a variety of electronic devices. However, the high cost of these panels hampers the mainstream entry of such devices. Nevertheless, non-stop research in this field instils the hope that the cost barrier will soon be mounted. The goal is to bring down the cost to a dollar per watt.

Another key criterion for solar cells is the efficiency of energy conversion. Today, efficiencies in the range of 12 to 18 per cent are possible, although most modules achieve only around 9 per cent in practice. Research therefore aims to improve this situation as well.

We covered a full range of innovations in solar cells in our tech focus story a year ago (EFY December 2009 issue). Here is a quick top-up thereafter.

Some companies are experimenting with novel desensitised solar cells

Light a billion lives

There are around a billion people in the world who do not have access to electricity and rely on oil lamps or firewood for basic illumination. Approximately 25 per cent of them live in India.

"Under the Light a Billion Lives (LaBL) programme, we will provide highly energy-efficient light-emitting diode (LED) lanterns to such areas and also solar charging options to run the same as they do not have access to electricity," says Hari Kiran of Sujana Energy, a member of the LaBL programme. The solar lanterns could be recharged at a village solar charging station, which will be rented out to the villagers at a nominal cost. These charging stations, while ensuring quality services at an affordable rate, will also provide green jobs to local people who manage these stations.

It has been found that each solar lantern saves 40-60 litres of kerosene per year, translating to Rs 100 billion burned each year in kerosene and wick lamps.

The environmental, social and economic benefits of the solar lantern make it a very commendable project by The Energy and Resources Institute (TERI).

using atomic layer deposition (ALD), which allows three-dimensional structures to be very thinly and uniformly coated, reducing the distance that charges travel within the device, consequently reducing the losses therein.

Others are trying to make solar panels themselves 'green'. While solar energy is green, some of the materials used in manufacturing solar panels are not necessarily so. For example, the main constituent of the back-sheets used in solar panels is petroleum. A company called BioSolar has come up with an alternative that uses a formulation based on cotton and a form of nylon resin made from castor bean oil. It provides a perfect moisture barrier as well as electric insulation, being an ideal material for the back-sheet.

Improvements and innovations on the production side also show promise. Thin-film solar cells based on cadmium telluride (CdTe) technology are gaining popularity. This technology greatly streamlines manufacturing and holds down cost while delivering competitive solar cell power. It is a very simple process where the semiconductor sheet is done in one shot, completely in vacuum. This is followed by laser-scribing and then module assembly.

A company called Applied Quantum Technology is trying to use the 'sputtering' process for the production of copper indium gallium (di)selenide (CIGS)-type thin-film photovoltaic cells. Sputtering is the proven standard for producing other high-volume cost-sensitive products

such as hard disks and optical disks. This approach includes single-step deposition, nano-engineering, device-enhancing source materials and simplified cell interconnect design.

Another company called XsunX Solar has come up with a new thin-film solar panel manufacturing technology that combines thermal evaporation with sputtering to create a new process that combines the high efficiency of thermal evaporation and the high throughput of sputtering. These systems take a 32-52 cm² (5-8 inch²) thin-film and process it the same way as it would a hard media disk. The hard disk drive technology lays down the back contact and then the evaporation system deposits the CIGS layer. Then subsequent processes lead to the final application of the front contact layer.

Ascent Solar has demonstrated the use of CIGS thin-film technology on a clear plastic substrate. Its cells are both extremely lightweight and flexible. The manufacturing process is done at the module level using roll-to-roll, monolithic integration. The process uses thermal co-evaporation to deposit the four metals—copper, indium, gallium and selenium—onto the moving polyvinyl, plastic substrate, resulting in an ideal CIGS compound with ideal thickness. Producing at the modular level means no external wiring. Everything is done internally just as in integrated circuit (IC) processing. The benefits of this are realised in large modules and at high voltages.